# Xiaohua Cai

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*Software Engineer with experience building full-stack web applications, REST APIs, authenticated workflows, and data-driven interfaces using React, Next.js, TypeScript, Node.js, PHP/Laravel, and relational databases. M.S. in Computer Science from Georgia Tech with a 4.0/4.0 GPA.*

## EDUCATION

**Master of Science in Computer Science**, Georgia Institute of Technology, GPA 4.0/4.0 2026
**Master of Science in Biotechnology**, University of Alberta, GPA 3.7/4.0 2018

## SKILLS

**Languages:** Python, Java, JavaScript, PHP, TypeScript, SQL, HTML/CSS/SASS
**Frameworks & Libraries:** React, Angular, Next.js, Node.js, Express, Laravel, Socket.io, Material UI, Tailwind CSS, Bootstrap
**Databases & APIs:** PostgreSQL, MySQL, MongoDB, Prisma, GraphQL, REST
**Infrastructure & Tools:** AWS, Docker, Docker Compose, Nginx, Git/GitHub, GitHub Actions, CI/CD, Postman, Ubuntu Linux

## WORK EXPERIENCE

**Software Development Engineer** December 2022 - June 2024
Affinity Bridge Inc.

- Developed React frontend features and PHP REST APIs for the NovaSafe application, a permission-intensive application with thousands of users; optimized database access through query restructuring, join/filter improvements, and targeted indexing, reducing SQL execution time by approximately 20% and implementing pagination, and reducing unnecessary rendering and full-page reloads.
- Built and shipped an internal application using Next.js, TypeScript, Prisma, and NextAuth; designed authenticated workflows, REST API endpoints, middleware, relational data models, and caching/revalidation strategies to reduce redundant data fetching and improve application performance.
- Built the Environmental Conventions Index with Next.js, developing interactive geographic maps with React Simple Maps and responsive scatter and line charts with ECharts(live site). Built reusable React UI components and custom hooks, integrated frontend workflows with REST APIs, and centralized shared data-fetching and state logic to reduce duplication and improve maintainability across the application.
- Built and maintained automated test coverage with Playwright and React Testing Library for critical user workflows and UI components, strengthening regression protection and reducing manual QA effort during releases;
- Supported production deployments and application releases by investigating build, configuration, API, and runtime failures and implementing fixes to improve application stability and release reliability.

**Software Development Engineer** January 2022 - December 2022
AMI RockChain Inc.

- Designed and implemented PHP Laravel REST APIs, PostgreSQL relational schemas, migrations, and CRUD workflows using MVC architecture; built authentication and role-based authorization to enforce access permissions across different user types and protected application workflows.
- Developed reusable, responsive UI components and user workflows with AngularJS, TypeScript, and Nebular UI, building maintainable production features across the application; Built interactive mapping and data-visualization features using Leaflet, Google Maps API, and ECharts, and integrated the Mailchimp API to support subscriber-management workflows.
- Deployed web applications to AWS, managed production releases, and diagnosed and resolved deployment issues.

**Software Development Engineer (Contract)** September 2021 - February 2022
Explorer Hop

- Built a full-stack internal web application using React, TypeScript, Material UI, Node.js, Express, and PostgreSQL, implementing REST APIs, backend business logic, and relational data workflows.
- Developed Shopify admin app features using React, Shopify App Bridge, and Polaris; integrated the Shopify GraphQL API and implemented webhooks for product-order events.

**Research and Development Scientist** January 2019 - August 2021
SBI BioEnergy Inc.

- Developed VBA automation tools to calculate mass balances for 10+ lab-scale reactors, reducing repetitive manual calculations and supporting experimental analysis; Analyzed, processed, and visualized experimental data using Python.

## PROJECTS

**Distributed Real-Time Chat Application** 2026
Tech stacks: React, Node.js, Express, Socket.IO, Redis, PostgreSQL, Nginx, Docker

- Re-architected a previously built single-server real-time chat application into a horizontally scalable distributed system, running multiple Socket.IO backend instances behind an Nginx load balancer.
- Implemented Redis Pub/Sub for cross-instance event propagation and Redis-backed shared room/presence state; persisted message history in PostgreSQL and containerized the distributed stack with Docker Compose.
- Built a concurrent WebSocket load-testing script and validated 200 simultaneous clients and 40,000 message deliveries with zero errors; verified continued availability for new connections after terminating one backend instance.